# Finding peers in a dynamic squad

**Multicast, unicast, C2-seeded, or CDS?**

**Choose how peers find each other.**
**Control topic audiences separately.**

CDS = Connext Cloud Discovery Service.

# Four choices, four operational trade-offs

| Strategy | Best fit | Main benefit | Main cost |
|---|---|---|---|
| Multicast | Multicast-capable LAN | No platform address roster | Network must deliver multicast |
| Static unicast | Stable, known deployment | Explicit contacts; no required central service | Repeated address/configuration maintenance |
| C2-seeded | Mission-controlled, changing squad | Central roster and application policy | Custom directory/update/recovery logic |
| Connext CDS | Dynamic peers; multicast unavailable | RTI-managed discovery rendezvous | Operate a reachable discovery service |

**Takeaway:** these establish discovery, not a user-data relay.
Endpoint matching and data delivery still happen between peers.

# Multicast: no roster, but the network must work

```
A announces to a shared group
B and C listen -> normal DDS discovery
```

## Gain

- Dynamic arrivals without maintaining every platform address.
- Simple on a multicast-capable local network.

## Give up

- Depends on multicast delivery across the actual network.
- Broad announcements may be undesirable over constrained radio links.

> **Takeaway:** verify multicast reachability first.
> Discovery multicast does not require user-data multicast.

# Unicast: explicit contacts, ongoing maintenance

```
A contacts B's configured discovery address
B accepts A -> normal DDS discovery
```

**Gain:** no multicast requirement and no mandatory central service.

**Give up:** address changes and arrivals require roster/configuration updates.
Copying every address into every platform multiplies maintenance.

A discovery address is configured as a DDS **peer descriptor**.
One-sided contact can work when the receiver accepts unknown peers.

> **Takeaway:** all-to-all configuration is a choice, not a DDS rule.

# C2: centralize the roster, not the data path

```
Platform -> C2: identity + discovery address
C2 -> platform: versioned directory
Platform -> peers: add new contacts, then discover directly
```

- Each platform starts with C2's address, not the whole squad roster.
- C2 distributes the peers needed for the mission and global services.
- Register explicit discovery addresses; user-data locator ports are not a substitute.
- Application policy chooses contacts; DDS `add_peer` applies them.

> **Takeaway:** C2 reduces configuration duplication.
> A full-squad directory does not automatically reduce discovery bandwidth.

# A teammate can repair a missed C2 update

```
P30 has C2 revision 12 and knows P31's address
P30 contacts P31 -> compares revisions -> shares C2-issued records
P31 validates the records and adds missing contacts
```

- Allow incoming discovery: `accept_unknown_peers=true`.
- Participant scopes and directory endpoints must still match.
- Cache/version records, avoid duplicate adds, and bound retries.
- A helper still needs a reachable address for the stranded platform.

> **Takeaway:** this is application directory repair, not built-in DDS gossip.
> Accepting unknown peers makes the directory a contact list, not an allowlist.

# CDS: RTI-managed discovery rendezvous

```
A / B -> CDS: participant announcements
CDS -> matching peers: forwarded announcements
A <-> B: direct endpoint discovery and user data
```

**Gain:** dynamic discovery without custom platform-directory distribution.
CDS can filter announcements by domain, tag, and participant partition.

**Give up:** service operations and a reachable bootstrap path.
Direct peer transport must still work; CDS is not Routing Service or a data broker.

7.7.0 SPDP2 mode forwards bootstrap only; configuration/liveliness stay peer-to-peer.
Configure compatible protocol modes and allow incoming peers.

> **Takeaway:** CDS supplies rendezvous; C2 can still own mission/team policy.

# Losing the seed is not the same as losing data

| Strategy | New peers or changed addresses | Existing peer communication |
|---|---|---|
| Multicast | Need multicast delivery or another seed | Depends on viable paths and normal DDS lifecycle |
| Static unicast | Need a reachable configured contact | Direct |
| C2-seeded | Need C2, cached records, or an informed teammate | Need not depend on C2 remaining available |
| CDS | Need CDS or an alternative bootstrap path | Documented to continue after discovery on viable direct paths |

**Takeaway:** central outage primarily threatens fresh discovery.
Restart, address change, lease expiry, or transport failure can still require rediscovery.

# Choose for operations. Measure traffic separately.

- **Multicast:** use where the network supports it.
- **Static unicast:** keep as a predictable baseline.
- **C2 directory + teammate repair:** reuse mission control and reduce roster maintenance.
- **CDS:** use RTI's rendezvous when avoiding custom bootstrap logic is the priority.

CDS and C2 policy can coexist; they solve different responsibilities.

Compare identical health, navigation, and team-data audiences under loss and restart.

> **Takeaway:** configuration simplicity is not a bandwidth result or authorization.

# Evidence and remaining gates

**Measured:** C2 add-once bootstrap, 6/6 isolated SPDP/SPDP2 runs.

**API-only:** Routing Service processor access, compiled/linked but not loaded.

**Documented CDS behavior:** introduction,
partition forwarding,
protocol modes,
service-outage tutorial.

**Still to test:** teammate repair, CDS deployment, loaded processor integration,
and comparable lossy-WAN traffic/convergence.

7.7.0 peer API documentation
limits `remove_peer` to SPDP. Do not assume SPDP2 peer removal or authorization semantics.